Econometrics Game - Case I

April 2026

Abstract

Novelty: Structural+Statistical+ML, studying benchmark models in the literature and extending them with improved performance, different horizons

We compare fundamental and statistical approaches to electricity price forecasting for short-term (1 week) and long-term (1 year) horizons. Using Danish data for 2018 to 2023, we estimate a fundamental model that forecasts load and renewables separately before inputting them into a price equation, and extend LEAR, a statistical model typically applied to day-ahead forecasting, to long-term forecasting. Our key finding is that the fundamental model is highly sensitive to fuel price assumptions. In contrast, LEAR provides better predictions across horizons. These results emphasize the trade-off between economic interpretability and forecasting robustness.

1 Introduction

Forecasting electricity prices accurately is a crucial problem for electricity market participants. Traders need to forecast prices at different intervals in order to make a profit. Electricity producers rely on forecasts to plan the development of new infrastructure, while grid planners need accurate forecasts to ensure system stability and limit costly backup electricity generation. In addition, governments and regulators need to know how prices will evolve to maintain the electricity sector's reliability and affordability.

However, forecasting electricity prices is a uniquely complicated issue, due to electricity being non-storable and the need for constant balance between supply and demand, leading to high volatility, seasonality, and periodicity in the price. The problem of obtaining accurate forecasts has been further complicated by the rise of renewables, particularly wind and solar. These have close to zero marginal cost, leading to prices that are close to zero or even negative, and exhibit high variability, increasing the volatility of electricity prices (Ketterer 2014, Ziel and Steinert 2018). Market liberalization since the 1990s and increased cross-border flows of electricity have added another layer of complexity, as electricity prices in one region depend also on supply and demand in interconnected regions. Furthermore, in recent years, energy crises have highlighted the importance of accurately predicting electricity prices so that governments can respond, at the same time as compounding the complexity of the forecasting problem (Segarra et al., 2024, Zakeri et al., 2023).

Different approaches have been taken to address this complex forecasting problem in the literature. Fundamental or structural models include economic factors, such as load, renewable production, fuel prices, and macroeconomic indicators, to explain prices using supply and demand reasoning (Weron, 2014). These models have the advantages of interpretability and a solid economic foundation, but typically perform worse in the short term due to underestimating volatility, and are sensitive to functional form specifications (Ghelasi and Ziel, 2025). Statistical models such as Lasso-Estimated AutoRegressive (LEAR) combine temporal price patterns with exogenous variables without explicitly modeling the economic relationships (Uniejewski et al., 2016). They are therefore more accurate for short-term predictions and are more robust to misspecification, although they cannot be interpreted as easily. A third class of approaches entails machine learning methods. These can effectively capture non-linear relationships, but are computationally demanding and lack transparency compared to other approaches (Ghelasi and Ziel, 2025).

In this paper, we address two research questions. Firstly, how can we best forecast spot prices for all hours for each day of the first week of 2024, using only data available up to the

end of 2023. Secondly, how can we best forecast the weekly average of hourly spot prices for every week of 2024, again using only data available up to the end of 2023. For both research questions, we seek to evaluate our forecasts with the actual observed spot prices for 2024. The time period of our data used for forecasting includes the European energy crisis, which we address in the paper.

Given the need to forecast across different horizons for these two research questions, we employ two different approaches to predict electricity prices. Firstly, we use a fundamental or structural model, explicitly modeling and predicting load, wind, and solar separately, and price as a function of these input variables. This approach is more relevant to our second research question concerning the longer time horizon. This approach also enables us to estimate coefficients on key factors determining the electricity price, including fuel prices. We extend this approach by considering alternative fuel price scenarios to test for sensitivity to fuel price assumptions. Secondly, we use a LEAR model which incorporates past prices as well as a rich set of exogenous variables. To answer our first research question, we extend the classic LEAR model used for day-ahead forecasting to multi-day ahead forecasting. We do not use the LEAR model to address our second research question as it is unsuited to longer time horizons. We evaluate both approaches against 2024 realizations using the root mean squared error (RMSE) and the mean absolute percentage error (MAPE). We also compare the predictions of our structural model and LEAR model for the first research question. In our evaluations, we focus on the accuracy of our predictions over short-term horizons, given the greater relevance to market participants.

PARAGRAPH: results PARAGRAPH: our contribution to the literature

We contribute to the literature by comparing two main approaches to forecasting electricity prices.

The rest of the paper is structured as follows. Section 2 provides descriptive statistics that inform our analysis. In Section 3, we describe our econometric approaches. Section 4 presents our key forecasting results, and evaluates these against the 2024 realizations. In Section 5, we discuss and interpret our results. Section 6 concludes.

Literature to add: https://www.sciencedirect.com/science/article/pii/S0140988325009442?viahttps://w (Ketterer 2014) https://www.sciencedirect.com/science/article/pii/S1364032118303885 (Ziel and Steinert 2018) https://ideas.repec.org/a/eee/intfin/v93y2024ics104244312400074x.html (Segarra et al. 2024) https://www.sciencedirect.com/science/article/pii/S2352484723013057 (Zakeri et al. 2023) https://www.sciencedirect.com/science/article/pii/S1364032125003570 (Ghelasi and Ziel 2025)

2 Data Description

This section presents descriptive statistics of key variables in the 2018-2023 dataset, documenting patterns that motivate the methodology chosen in Section 3. We establish four key empirical facts: a negative relationship between renewable generation and electricity prices, a positive relationship between load and electricity prices, high autocorrelation between electricity prices, and high variability, seasonality, and periodicity in key variables.

Figure 1 shows that higher combined renewable generation forecasts are negatively associated with electricity prices in DK1 over 2018–2023 (correlation: $r = -0.210$). When renewable output is high, cheap generation displaces more expensive sources and the marginal cost of electricity falls, driving prices down. This is referred to as the merit-order effect, and is a key economic mechanism in our analysis. We also note that the relationship appears largely linear.

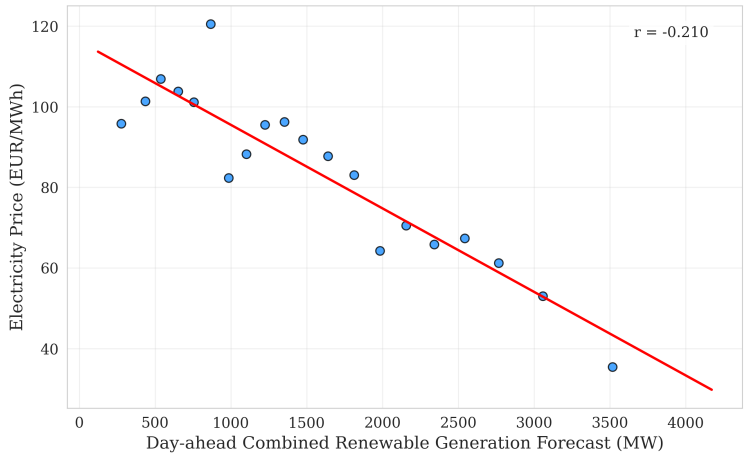


Figure 1: Binscatter of day-ahead combined renewable generation forecast and electricity prices, DK1, 2018–2023

Notes: Each point represents a bin of hourly observations of the day-ahead combined renewable generation forecast and day-ahead electricity spot prices for DK1 over 2018–2023. The combined renewable generation forecast is constructed as the sum of the day-ahead onshore wind, offshore wind, and solar generation forecasts.

Figure 2 shows the relationship between total load and electricity prices in DK1 over 2018–2023. The correlation is positive and statistically significant ($r = 0.058$, $p = 0.006$). Higher demand requires using more expensive generation sources, pushing prices up due to the merit order. The strong positive relationship between load and electricity prices and strong negative relationship between renewables and prices motivate our fundamental model in Section 3.

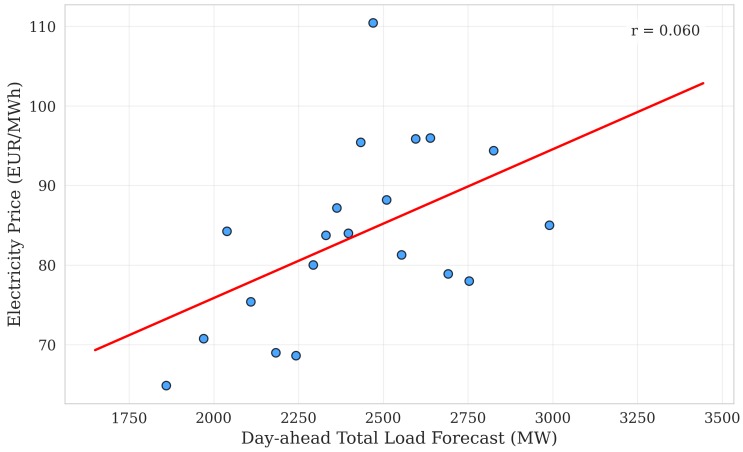


Figure 2: Binscatter of day-ahead load and electricity prices, DK1, 2018–2023

Notes: Each point represents a bin of hourly observations of total load and electricity spot prices for DK1 over 2018–2023.).

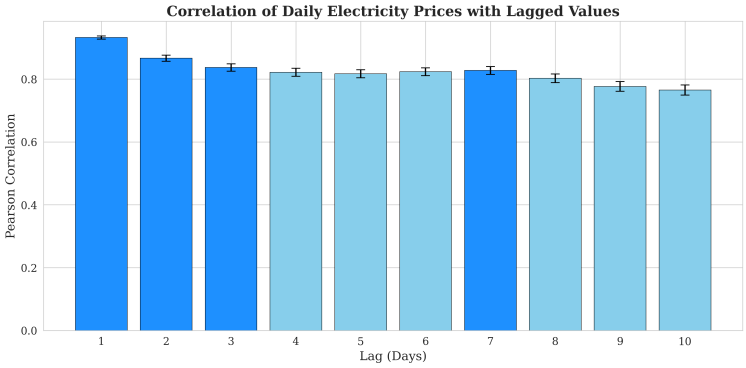


Figure 3: Electricity Price Lag Correlations

Notes: The figure is based on hourly day-ahead electricity spot prices for Denmark (DK1) from 2018–2023. Lag correlations are computed using historical spot prices.

Figure 3 shows the Pearson correlation between daily electricity prices and their lagged values for up to 10 days. The correlation is highest at lag 1 (0.93), declining to 0.87 at lag 2 and 0.84 at lag 3. Lags 4 through 7 are similar at around 0.82–0.83. This high autocorrelation informs our LEAR model in Section 3.

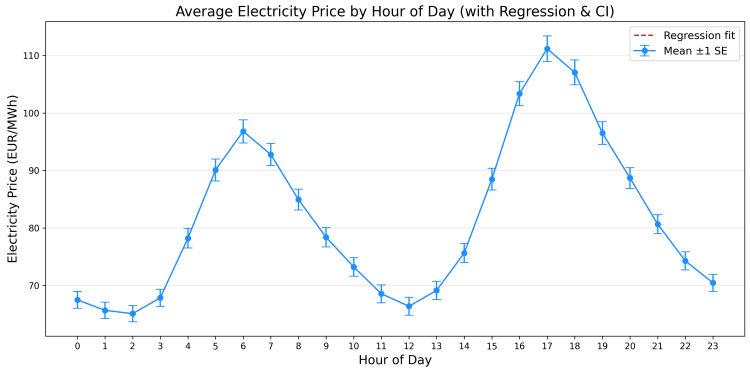


Figure 4: Average daily electricity price by hour of the day, DK1, 2018–2023

Notes: Average hourly day-ahead electricity spot prices for DK1 over 2018–2023, computed separately for each hour of the day.

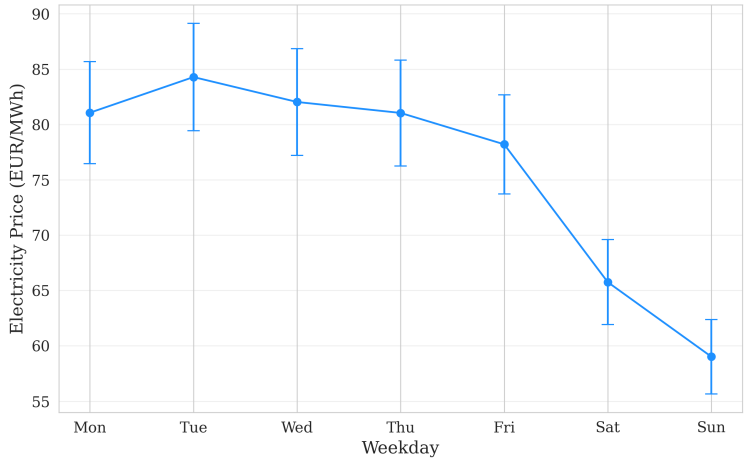


Figure 5: Average daily electricity price by day of the week, DK1, 2018–2023

Notes: Average hourly day-ahead electricity spot prices for DK1 over 2018–2023, computed separately for each day of the week. Error bars indicate 95% confidence intervals.

Next, we document high variability, periodicity, and seasonality of key variables. Figure 4 shows high hourly periodicity in electricity prices. Prices peak at 06:00 and again at 17:00. Prices are lowest at 02:00 and fall again at 12:00. These trends likely reflect that load is lowest at night, while renewable generation peaks at around midday.

Figure 5 shows the average daily electricity price by day of the week. Prices are broadly stable from Monday to Thursday, ranging between 81 and 84 EUR/MWh, before declining sharply from Friday onward. Saturday and Sunday average 65.9 and 59.2 EUR/MWh respectively, roughly 22 EUR/MWh below the midweek level. This reflects a systematic reduction in industrial and commercial electricity demand over the weekend, which reduces reliance on generation by high marginal cost sources.

Public holidays are also associated with substantially lower electricity prices relative to regular working days, as shown in Figure 6. Average prices on holidays are 53.2 EUR/MWh compared to 76.6 EUR/MWh on non-holidays, a gap of approximately 30%. This reflects a sharp reduction in industrial and commercial demand on public holidays.

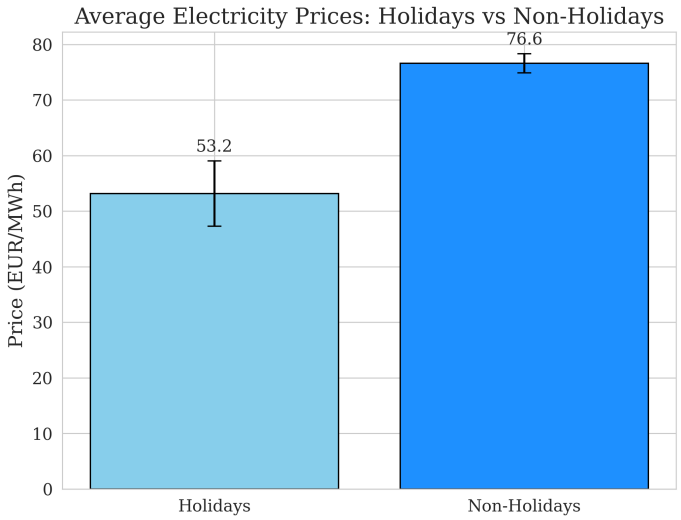


Figure 6: Average electricity prices on public holidays versus non-holidays, DK1, 2018–2023

Notes: Average hourly day-ahead electricity spot prices for DK1 over 2018–2023, separately for public holidays and non-holiday days. Error bars indicate 95% confidence intervals.

Figure 7 illustrates several patterns in load, wind, and solar forecasts. We see an increase in forecasted load over the time period, suggesting a linear long-run time trend. We also see that load is highly seasonal, with a sharp decrease in July, likely reflecting holidays. We also see an increase in forecasted wind, possibly reflecting a capacity expansion over time, which seems largely linear. Wind also exhibits a highly volatile seasonal component, which is typically higher in winter months. Solar also exhibits an increase over time, also possibly due to capacity expansion, although this increase seems non-linear. Solar is also seasonal, with much higher values in summer months compared to winter.

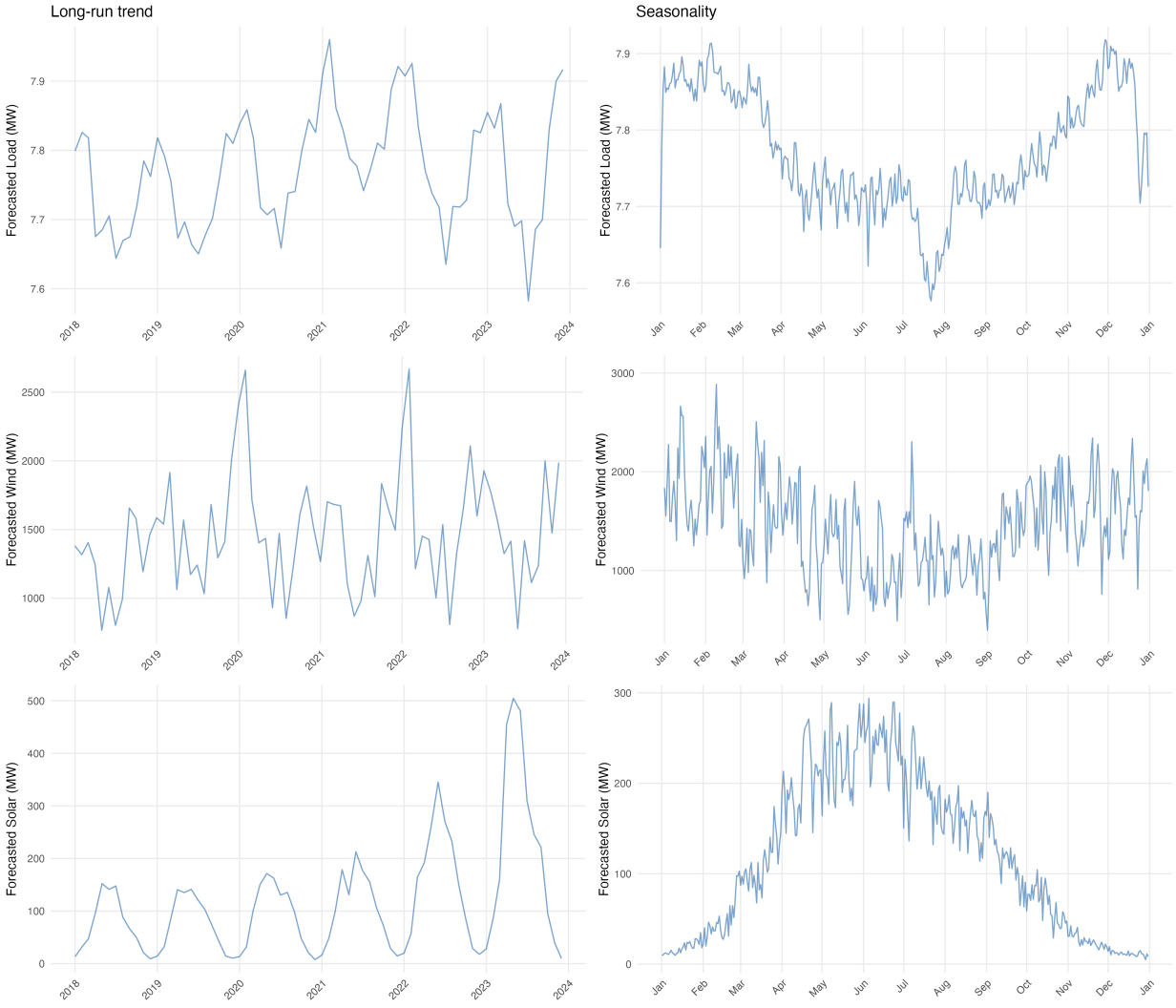


Figure 7: Load, Wind and Solar Forecasts: Long-run Trend and Seasonality, DK1, 2018–2023

Notes: Long-run trends show averages over hours and days for each month and year. Seasonality displays averages over hours and years for each month and day.

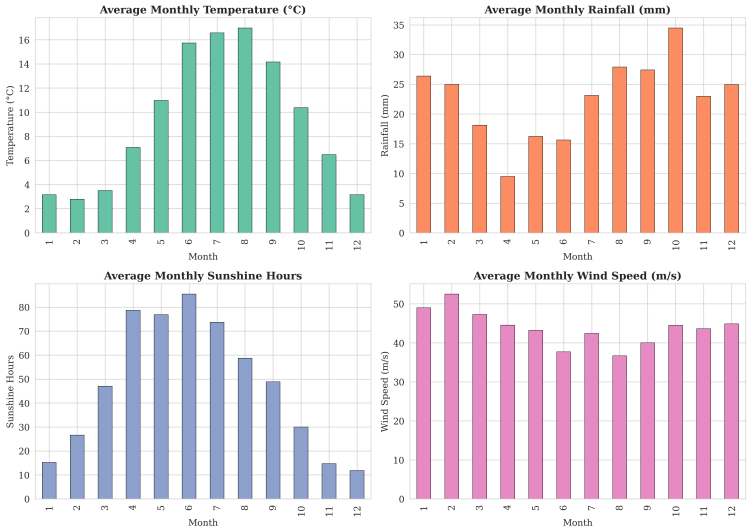


Figure 8: Seasonal trends in climate variables, DK1, 2018–2023

Notes: Monthly averages of daily climate data for Denmark (DK1) over 2018–2023. Variables include mean temperature (°C), total precipitation (mm), sunshine duration (hours), and mean wind speed (m/s), aggregated across weather stations assigned to DK1. Only observations passing ECA&D quality control are included.

Figure 8 shows that climate variables are highly seasonal. Temperature and sunshine duration are highest in June–August and fall through the winter, while wind speed peaks in autumn and winter. Precipitation is higher in autumn and lower in spring. These patterns have implications for electricity prices. In winter, wind speeds increase renewable generation, pushing electricity prices down. In summer, wind is weaker and solar takes over as the main renewable energy source, decreasing electricity prices. Seasonal climate variation is therefore likely one of the primary drivers of electricity price seasonality, through affecting load and the generation of renewables.

In the following section, we describe our methods for forecasting electricity prices, building on the descriptive findings presented in this section.

3 Methodology

3.1 Fundamental Model

We first employ a fundamental model approach. The economic intuition is that electricity market clearing prices are determined by forecasted supply and demand. Under the merit order, price is driven by residual load (the difference between demand and renewable supply) as well as fuel and carbon prices. The approach consists of two stages. First, we model

forecasted load, wind, and solar generation as functions of seasonal, periodical, and time-trend components, motivated by the observed patterns in 2018–2023. This allows us to forecast these fundamentals for 2024. Second, we estimate a reduced-form price model on 2018–2023 data, obtaining coefficients on the forecasted fundamentals, fuel prices, and carbon prices. We then input our 2024 forecasts of fundamentals into this price model to generate 2024 price forecasts.

Forecasted Load Demand Forecasted load demand is driven by hour of the day, trend, business-cycle, and seasonality, holiday as well as day-of-week patterns. The decision for specification are based on the observed trends and seasonality in Figure 7. The following equation is estimated on 2018-2023 data:

$$\ln(\text{Load}_{d,h}) = \alpha_h^L + \beta_h^L \cdot t + \gamma_h^{L,t} \sin\left(\frac{2\pi t}{365.25 * 4}\right) + \delta_h^{L,t} \cos\left(\frac{2\pi t}{365.25 * 4}\right)$$

$$+ \sum_{k=1}^{3} \left[\gamma_{h,k}^L \sin\left(\frac{2\pi k d}{365.25}\right) + \delta_{h,k}^L \cos\left(\frac{2\pi k d}{365.25}\right)\right] + \text{DOW}_d + \lambda_h^L \cdot \text{SummerBreak}_d + \mu_h^L \cdot \text{WinterBreak}_d + \varepsilon_{d,h}^L$$

where t denotes a linear time trend (days since Jan 1, 2018). Inspired by Figure 7 the long-run demand is modeled as a linear trend and a 4 year business cycles using $\sin(\cdot), \cos(\cdot)$. To capture seasonality, we include $\sin(\cdot), \cos(\cdot)$ of frequencies 1 to 3 in the day d of the year. Summer break is a dummy for July and winter break for the last two weeks of December. To account for variation in load demand over the week we include day-of-week dummies DOW_d.

Forecasted Wind Generation Forecasted wind production depends on hour of the day, seasonal patterns and a linear capacity expansion trend:

$$\ln(\text{Wind}_{d,h}) = \alpha_h^W + \beta_h^W \cdot t + \sum_{k=1}^{6} \left[\gamma_{h,k}^W \sin\left(\frac{2\pi k d}{365.25}\right) + \delta_{h,k}^W \cos\left(\frac{2\pi k d}{365.25}\right)\right] + \varepsilon_{d,h}^W. \quad (1)$$

As shown in Figure 7 the seasonal pattern is more complex than a simple sinusoid. Thus, we approximate it using a Fourier series up to the sixth harmonic in d. Visual plotting shows that higher harmonics lead to overfitting. At the same time, expansion appears to be linear in t.

Forecasted Solar Generation Forecasted solar production is a function of hour of the day, seasonality, and capacity expansion:

$$arcsinh(\text{Solar}_{d,h}) = \alpha_h^S + \beta_{1,h}^S \cdot t + \beta_{2,h}^S \cdot t^2 + \sum_{k=1}^{3} \left[\gamma_{h,k}^S \sin\left(\frac{2\pi k d}{365.25}\right) + \delta_{h,k}^S \cos\left(\frac{2\pi k d}{365.25}\right) \right]$$

$$+ \sum_{k=1}^{2} \left[\tilde{\gamma}_{h,k}^S \sin\left(\frac{2\pi k d}{365.25}\right) \cdot t + \tilde{\delta}_{h,k}^S \cos\left(\frac{2\pi k d}{365.25}\right) \cdot t \right] + \varepsilon_{d,h}^S$$

As depicted in Figure 7 capacity expands with a linear and quadratic trend in t. Seasonality is less complex and modeled using a Fourier series up to the third harmonic in d. However, variation between hours and seasonality both increase with capacity expansion, which is modeled using interactions. To handle zero values during night, solar production is transformed to its area hyperbolic sine.

3.2 Price Forecasting using Structural Model

The electricity market clears when demand equals supply. The residual load is covered by other sources, ordered by marginal cost (the merit order):

$$\text{Residual}_{d,h} = \text{Load}_{d,h} - \text{Wind}_{d,h} - \text{Solar}_{d,h} \tag{2}$$

The spot price is determined by the marginal cost of the last (most expensive) source. This cost depends on $\text{Residual}_{d,h}$, fuel prices, and carbon prices.

3.2.1 Reduced-Form Price Model

We estimate a separate regression for each hour of the day using all 2018-2023 data:

$$\begin{aligned} \text{Price}_{d,h} = \alpha_h + \beta_h \cdot \text{Load}_{d,h} + \gamma_h \cdot \text{Wind}_{d,h} + \delta_h \cdot \text{Solar}_{d,h} + \zeta_h \cdot \text{GasPrice}_{d-1} \\ + \iota_h \cdot \text{OilPrice}_{d-1} + \eta_h \cdot \text{CoalPrice}_{d-1} + \theta_h \cdot \text{CarbonPrice}_{d-1} + \varepsilon_{d,h} \end{aligned} \tag{3}$$

where α_h is an hour-specific intercept, $\beta_h > 0$ is the coefficient on load, $\gamma_h < 0$ the coefficient on wind, $\delta_h < 0$ the coefficient on solar, $\zeta_h, \iota_h, \eta_h > 0$ are fuel price sensitivities, and $\theta_h > 0$ is the carbon price sensitivity. We lag the fuel prices by one day to reflect when the day-ahead market clears.

3.3 Day-Ahead Hourly Forecasts

For each (day, hour) pair, we generate point forecasts. Since 2024 fuel prices and climate data are not available, we use 2023 realizations as a proxy for 2024 to capture recent seasonal patterns.

$$\widehat{\text{Price}}_{d,h}^{2024} = \hat{\alpha}_h + \hat{\beta}_h \cdot \widehat{\text{Load}}_{d,h}^{2024} + \hat{\gamma}_h \cdot \widehat{\text{Wind}}_{d,h}^{2024} + \hat{\delta}_h \cdot \widehat{\text{Solar}}_{d,h}^{2024}$$
$$+\hat{\zeta}_h \cdot \text{GasPrice}_{d-1}^{2023} + \hat{\iota}_h \cdot \text{OilPrice}_{d-1}^{2023} + \hat{\eta}_h \cdot \text{CoalPrice}_{d-1}^{2023} + \hat{\theta}_h \cdot \text{CarbonPrice}_{d-1}^{2023} \quad (4)$$

where $\widehat{\text{Load}}_{d,h}, \widehat{\text{Wind}}_{d,h}, \widehat{\text{Solar}}_{d,h}$ are forecasted fundamentals, and $\hat{\alpha}_h, \hat{\beta}_h, \ldots$ denote OLS estimates from the reduced-form price model.

3.3.1 Average Hourly Forecasts Per Week

Once we have forecasts for all 8,760 (day, hour) pairs, we compute the weekly average for each hour:

$$\widehat{\text{Price}}_{w,h} = \frac{1}{7} \sum_{d \in \text{ week } w} \widehat{\text{Price}}_{d,h} \quad (5)$$

3.4 Extensions and Robustness

3.4.1 Sensitivity to Fuel Price Assumptions

We estimate forecasts under alternative fuel price scenarios. Our baseline (scenario 1) uses fuel prices from 2023 as a proxy for 2024. However, we hypothesize that forecasts will be highly sensitive to fuel prices as these largely determine the marginal cost of the residual load. We therefore examine three further fuel price scenarios as proxies for 2024: pre-energy crisis (2018-2021) fuel prices (scenario 2); 20% increase/decrease on 2023 prices (scenario 3); and 2022 energy crisis fuel prices (scenario 4).

3.5 LEAR Model

The Lasso-Estimated AutoRegressive (LEAR) model of Lago et al. (2021), originally introduced as *LassoX* by Uniejewski et al. (2016), is a benchmark in day-ahead electricity price forecasting. We first present the benchmark LEAR specification for day-ahead forecasting, then extend it to 7-day ahead hourly forecasting with additional predictors to improve predictive performance.

For the day-ahead forecast, LEAR estimates a separate model for each hour $h \in \{1, \ldots, 24\}$, yielding 24 independently estimated LASSO regressions:

$$
\begin{aligned}
p_{d,h} = & \underbrace{\sum_{i=1}^{24} \theta_{h,i}\, p_{d-1,i} + \sum_{i=1}^{24} \theta_{h,24+i}\, p_{d-2,i} + \sum_{i=1}^{24} \theta_{h,48+i}\, p_{d-3,i} + \sum_{i=1}^{24} \theta_{h,72+i}\, p_{d-7,i}}_{\text{96 autoregressive terms}} \\
& + \underbrace{\sum_{i=1}^{24} \theta_{h,96+i}\, x^1_{d,i} + \sum_{i=1}^{24} \theta_{h,120+i}\, x^2_{d,i} + \sum_{i=1}^{24} \theta_{h,144+i}\, x^1_{d-1,i} + \sum_{i=1}^{24} \theta_{h,168+i}\, x^2_{d-1,i}}_{\text{96 exogenous terms: days } d \text{ and } d-1} \\
& + \underbrace{\sum_{i=1}^{24} \theta_{h,192+i}\, x^1_{d-7,i} + \sum_{i=1}^{24} \theta_{h,216+i}\, x^2_{d-7,i}}_{\text{48 exogenous terms: day } d-7} + \underbrace{\sum_{i=1}^{7} \theta_{h,240+i}\, z_{d,i}}_{\text{7 day-of-week dummies}} + \varepsilon_{d,h}
\end{aligned}
\tag{6}
$$

Following (Lago et al., 2021), we add three blocks of regressors in the model:

- **Autoregressive terms (96 variables):** All 24 hourly prices from lag days $d-1$, $d-2$, $d-3$, and $d-7$ are included. Lags d-4 to d-6 are excluded as they fall on different weekdays and have low predictive power.
- **Exogenous terms (144 variables):** Following Lago et al. (2021), two exogenous variables are included at all 24 hours for days d, $d-1$, and $d-7$, yielding $2 \times 3 \times 24 = 144$ regressors. x^1 is the day-ahead total load forecast, capturing demand-side. x^2 is the combined day-ahead renewable generation forecast (onshore wind, offshore wind, and solar). Higher renewable generation decreases prices.
- **Day-of-week dummies (7 variables):** A binary vector $\mathbf{z}_d = [z_{d,1}, \ldots, z_{d,7}]^\top$ for the each day of the week (Lago et al., 2021).

LASSO Estimation

Given the large number of regressors, OLS would suffer from overfitting. Following Lago et al. (2021), we therefore estimate parameters via LASSO (Tibshirani, 1996):

$$\hat{\boldsymbol{\theta}}_h = \arg\min_{\boldsymbol{\theta}_h} \left\{ \sum_{d=8}^{N_d} \left(p_{d,h} - \hat{p}_{d,h}\right)^2 + \lambda \sum_{i=1}^{247} |\theta_{h,i}| \right\} \tag{7}$$

The ℓ_1 penalty shrinks coefficients of irrelevant regressors to exactly zero i.e. performing automatic variable selection among the 247 candidate predictors. Following Lago et al. (2021), λ is selected daily via LARS with in-sample AIC and the model is re-estimated at the optimal λ using coordinate descent.

Variance-Stabilizing Transformation

Electricity prices exhibit heavy tails and frequent spikes. Following Lago et al. (2021), all variables are transformed prior to estimation using the area hyperbolic sine (asinh) transformation:

$$\text{asinh}(x) = \log\left(x + \sqrt{x^2 + 1}\right) \tag{8}$$

where x is the price standardized by the in-sample median and median absolute deviation. Unlike the log transformation, asinh handles zero and negative prices, which occur frequently in our data. Predictions are back-transformed to EUR/MWh after forecasting.

Calibration Windows

The model is re-estimated daily on a rolling window of W days. A short W would work better at capturing recent market developments but may be unstable because of the few observations given the high number of predictors. Following Lago et al. (2021), we estimate the model under four calibration windows: 8 weeks, 52 weeks, 3 years, and 4 years and average the resulting forecasts. Lago et al. (2021) show this approach consistently outperforms any single window across European electricity markets.

3.6 Extending LEAR for (Hourly) Multi-Day Ahead Forecasting

To forecast all seven days of the first week of January 2024 using only data up to 31 December 2023, we iterate LEAR forward day by day. Unavailable lagged prices are replaced by previously generated LEAR forecasts at each iteration, meaning that each day's prediction feeds directly into the next. Day-ahead exogenous variables are only available for January 1

from the December 31 auction. We consider two variants for this problem. *Baseline LEAR* drops missing regressors from the specification. *Structural LEAR* replaces missing exogenous variables with predictions from the predictions obtained from 3.1.

3.7 Extending LEAR: Additional Variables

The baseline LEAR specification of Lago et al. (2021) includes day-ahead load and combined renewable generation as the two exogenous variables. We augment this with four additional predictors. First, Danish bank holiday dummies capture systematic price deviations on public holidays, which exhibit demand profiles closer to Sundays than regular weekdays (called holidays). Second, we replace the combined renewable forecast with separate onshore wind, offshore wind, and solar forecasts (*split renewables*), allowing the model to distinguish the distinct price effects of each source. Third, cross-border transmission flows between DK1 and neighbouring price zones are included, (called channel transmission) capturing the effect of import and export constraints on local prices. Fourth, daily energy commodity prices – natural gas, oil, coal, and EUA carbon (called energy) – enter as additional exogenous variables, reflecting fuel cost conditions at the time of the auction.

All additional exogenous variables enter at lags $d-1$ and $d-7$, following the lag structure of the baseline LEAR specification. For the multi-day ahead forecasts, these variables are not predicted – if they fall within the forecast period and are therefore unobserved, they are dropped from the specification for that day, consistent with the *Baseline LEAR* approach. They always enter the day-ahead forecast for January 1, where all $d-1$ values are observed.

Natural gas prices are included because gas-fired generation is the marginal technology in DK1 whenever wind output is insufficient to meet demand. Load and renewable forecasts capture how much thermal generation is required, but carry no information about how expensive that generation will be. Gas price fills this gap: on days when gas costs 50 EUR/MWh versus 150 EUR/MWh, electricity prices will differ substantially even under identical load and wind conditions. EUA prices are included because carbon allowances represent a direct additive cost on top of fuel costs. Every megawatt-hour from a thermal plant requires surrendering allowances, so EUA price movements translate one-for-one into changes in the marginal cost of generation, independently of load and renewable conditions (Weron, 2014). Both variables are included at lags $d-1$ and $d-2$, with LASSO determining

which lags carry genuine predictive content.[1]

3.8 Weekly Price Forecasting

We extend the LEAR model of Lago et al. (2021) to long-horizon weekly average price forecasting. We adapt the model structure to the weekly horizon, following the long-term profile forecasting framework of Wagner et al. (2022).

3.8.1 Estimation Sample

The model is estimated on the full historical sample, comprising weekly average day-ahead prices for the DK1 price zone from week 1 of 2018 through week 52 of 2023, yielding $N = 312$ weekly observations.

3.8.2 Model

Let p_w denote the average day-ahead spot price in week $w \in \{1, \ldots, 52\}$. The model is:

$$\begin{aligned} p_w = a_0 + \sum_{k=1}^{K} \left[a_k \sin\left(\frac{2\pi k w}{52}\right) + b_k \cos\left(\frac{2\pi k w}{52}\right) \right] \\ + \beta_1 \overline{\text{gas}}_w + \beta_2 \overline{\text{co2}}_w + \beta_3 \overline{\text{wind}}_w + \beta_4 \overline{\text{temp}}_w + \beta_5 \overline{\text{ecpi}}_w + \beta_6 \overline{\text{gdp}}_w + \beta_7 \overline{\text{stoxx}}_w \\ + \phi_1 p_w^{2023} + \phi_2 p_w^{2022} + \varepsilon_w, \end{aligned} \tag{9}$$

where all parameters are estimated jointly via LASSO with the regularization parameter λ selected by expanding-window time-series cross-validation.

Seasonal component. The terms $\sin(2\pi kw/52)$ and $\cos(2\pi kw/52)$ capture the annual seasonality as a function of the week-of-year w. Each k adds one additional cycle per year: $k = 1$ captures the annual cycle driven by heating demand, and $k = 2$ the semi-annual cycle. Both sin and cos terms are needed for each k so that the phase (the precise week of the

[1] Coal prices, oil prices, macro-financial variables (VIX, EuroSTOXX50, MSCI, EUR/USD), CPI, GDP, and transmission flows are excluded. Coal is an infrequent price-setter in DK1 given limited domestic coal capacity, and its marginal contribution is likely absorbed by the gas price term. Oil has no direct link to electricity generation in DK1. The macro-financial variables reflect financial market sentiment rather than physical generation costs and provide no direct mechanism through which they would predict hourly spot prices. CPI and GDP operate at monthly and quarterly frequency respectively, making them incompatible with hourly price dynamics. Transmission data in the dataset contains realized cross-border flows rather than day-ahead capacity announcements; their predictive content is largely absorbed by the lagged price terms already present in LEAR.

seasonal peak) is determined by the data. The parameters $\{a_k, b_k\}$ are estimated from the 2018–2023 sample and evaluated at $w = 1, \ldots, 52$ for the 2024 forecast (Wagner et al., 2022). $K \in \{1, 2, 3\}$ is selected by the Akaike Information Criterion on the estimation sample.

Fundamental component. All regressors are constructed to be comparable across years. $\overline{\text{gas}}_w$, $\overline{\text{co2}}_w$, $\overline{\text{wind}}_w$, $\overline{\text{temp}}_w$, $\overline{\text{ecpi}}_w$, $\overline{\text{gdp}}_w$, and $\overline{\text{stoxx}}_w$ are weekly averages, normalized by their yearly mean in year y containing week w:

$$\overline{x}_w = \frac{x_w}{\bar{x}_y}, \tag{10}$$

where $\bar{x}_y$ denotes the yearly mean of x in year y. This normalization removes structural level shifts across years – particularly relevant for the fuel cost and macro variables given the energy crisis within the estimation sample – so that all regressors capture relative deviations from the annual norm rather than absolute levels. Gas is the marginal fuel in the DK1 merit order during periods of low renewable output; carbon costs pass through directly to electricity prices via the marginal cost of generation. $\overline{\text{wind}}_w$ and $\overline{\text{temp}}_w$ are normalized weekly averages of wind production and mean temperature from the renewable production and climate datasets, capturing week-specific variation in renewable supply and heating demand. The business cycle variables $\overline{\text{ecpi}}_w$, $\overline{\text{gdp}}_w$, and $\overline{\text{stoxx}}_w$ capture aggregate demand conditions that affect electricity consumption at longer horizons. For the 2024 forecast, all variables are set to their week-w values from 2023, normalized by the 2023 yearly mean.

Autoregressive component. p_w^{2023} and p_w^{2022} are the realized prices in week w of 2023 and 2022 respectively, normalized by the yearly mean. Both lags are observed before the forecast period begins. These are the only feasible autoregressive lags at the weekly horizon: all lags shorter than 52 weeks fall within the forecast period and are unavailable at forecast time. Normalizing by the yearly mean ensures that the AR lags capture relative week-of-year price patterns, i.e., whether a given week is historically expensive or cheap relative to its year, rather than absolute price levels that shift dramatically across years due to structural breaks such as the energy crisis. They mirror the use of multiple weekly lags in LEAR (Lago et al., 2021) and are within the estimation window for our sample.

3.8.3 Estimation

Collecting all regressors into a design matrix $\mathbf{X}$ of dimension $312 \times P$ and stacking the 312 weekly observations, the LASSO estimator is:

$$\hat{\boldsymbol{\theta}}_\lambda = \arg\min_{\boldsymbol{\theta}} \left\{ \frac{1}{312} \|\mathbf{p} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}_{-0}\|_1 \right\}, \tag{11}$$

where $\boldsymbol{\theta}_{-0}$ excludes the intercept from penalization and all regressors are standardized to zero mean and unit variance prior to estimation. With $N = 312$ observations and at most $P = 2K + 9$ regressors, LASSO regularization prevents overfitting and performs automatic variable selection among potentially correlated regressors.

3.8.4 Benchmark

As a benchmark we implement the sinusoidal model of Wagner et al. (2022), which retains only the Fourier seasonal component of equation (9) estimated by OLS, with no fundamental or autoregressive regressors. This allows us to isolate the marginal contribution of the fundamental and autoregressive components to forecast accuracy relative to a pure seasonality baseline.

4 Empirical Results

Table 1: Forecast Errors by Day — First Week of 2024

Scenario	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	RMSE	Rel.RMSE
Mean price (EUR/MWh)	33.50	52.36	48.31	86.61	91.21	88.94	87.07	69.72	—
Baseline (2023)	100.26	66.46	74.97	49.31	46.03	44.63	65.80	66.51	95.4%
Good case (2023 ×0.8)	75.37	44.44	52.01	25.25	24.88	25.21	43.64	44.97	64.5%
Bad case (2023 ×1.2)	125.31	89.96	98.85	74.05	68.96	65.54	88.49	89.39	128.2%
2022 prices	117.30	59.98	67.09	41.16	57.03	78.61	99.62	78.29	112.3%
Avg 2018–2021	15.07	45.80	37.27	53.52	53.71	44.46	32.11	42.21	60.5%
Dec 2023 prices	35.71	24.11	23.33	19.17	20.30	14.53	15.17	22.75	32.6%

4.1 Fundamental Model

We also observe that the results are highly sensitive to fuel price assumptions.

Table 2: Forecast Errors by Quarter — Weekly Predictions 2024

Scenario	Q1	Q2	Q3	Q4	RMSE	Rel.RMSE	MAE	Rel.MAE
Mean price (EUR/MWh)	65.01	60.49	66.15	88.07	69.86	—		—
Baseline (2023)	38.46	30.65	18.92	25.40	29.09	41.6%	24.07	34.5%
Good case (2023 ×0.8)	20.66	19.06	14.74	28.64	21.27	30.4%	16.83	24.1%
Bad case (2023 ×1.2)	56.65	42.74	27.74	29.63	40.67	58.2%	35.34	50.6%
2022 prices	114.41	125.87	275.71	135.97	178.08	254.9%	158.47	226.8%
Avg 2018–2021	33.78	14.41	19.64	30.87	25.82	37%	21.47	30.7%
Dec 2023 prices	12.19	27.57	18.37	26.34	21.96	31.4%	17.69	25.3%

4.2

5 Discussion and Economic Interpretation

6 Conclusion

Electricity price forecasting is a complex problem which has received considerable attention in the literature, and spawned a variety of approaches. In this

We first document that

We next investigate how two main approaches compare in forecasting prices over short- and long-term horizons. In line with our first hypothesis, the LEAR model performs better in the short-term. This likely reflects that in the short-term, autocorrelation plays an important role. Contrary to our second hypothesis, our structural model does not perform better in the

Our structur